

● SATELLITE OUTAGE PROTECTION, AUTOMATED

# *SpaceShield*

Autonomous SLA enforcement for satellite internet. When a satellite goes dark, compensation doesn't wait for a human — it's detected, cryptographically proven, and paid out in about fifteen seconds.

Spacecoin

Attestcoin

Creditcoin

# Satellite internet fails. Compensation doesn't happen.

## Manual claims

Slow, costly, and requires someone who just lost connectivity to file a report — exactly when they can least afford to.

## Centralized refunds

Someone has to decide what counts as a "real" outage. That someone is a trust bottleneck DePIN was supposed to remove.

## No compensation at all

The default outcome. The user absorbs 100% of the downtime risk on infrastructure they never controlled.

# No human decides whether an outage happened. No human decides who gets paid.

## 01 · DETECT

### AI Agent

Watches satellite status and cross-checks it against an independent public tracker (real NORAD/CelesTrak data) before it will trigger anything.



## 02 · VERIFY

### Attestcoin

The native Block Prover precompile cryptographically checks a Merkle + continuity proof. Not an opinion — a proof.



## 03 · SETTLE

### Creditcoin

Pays out automatically from the operator's pre-locked bond, pro-rated to how long each subscriber was actually covered.

# A pull, not a push — nobody's money moves without them asking.

- 1 An operator locks a bond on Creditcoin — collateral that guarantees the money to pay claims actually exists.
- 2 A subscriber locks stake into SpaceShield's own coverage vault to get protected.
- 3 An outage is detected and independently cross-checked before anything moves.
- 4 Multiple independent oracles each verify the outage through Attestcoin and must agree — no single party can finalize a payout alone.
- 5 Compensation is calculated pro-rata (proportional to coverage time vs. outage length) and becomes *claimable* — each subscriber pulls their own share, directly to their wallet.

NOT VAPORWARE

# What's actually live today

| PIECE                         | STATUS  | EVIDENCE                                                                           |
|-------------------------------|---------|------------------------------------------------------------------------------------|
| Contract logic                | Live    | 16/16 tests passing – bonding, pull claims, pro-rata, M-of-N attestation           |
| Attestcoin precompile         | Live    | Real rejection from 0x0FD2 on Creditcoin testnet: "Merkle proof validation failed" |
| Creditcoin testnet deployment | Live    | Chain 102031 – 4 contracts deployed, addresses on next slide                       |
| Oracle Worker                 | Live    | Always-on, independently re-verifies every attestation before signing              |
| AI detection agent            | Live    | Real Celestrak / NORAD satellite-tracking cross-check                              |
| Satellite telemetry           | Pending | Waiting on Spacecoin – see "What's left"                                           |

LIVE ON CREDITCOIN CC3-TESTNET

# Real contracts. Real addresses. Real transactions.

| CONTRACT            | ADDRESS                                    |
|---------------------|--------------------------------------------|
| MockSpacecoinSource | 0xBDBFCFa52d153255CFC64a18e552ff826c53c0B7 |
| CoverageVault       | 0xC1Db5E73A139c6Fd4Eac6321ee31F7876d8f1d34 |
| SettlementContract  | 0xF5F31F256Dcd5b27f480600F8B660D936Be4c4Ec |
| SpaceShieldASC      | 0xC1d9eA6DeDD6C1f1290220DC91fbC8fD2b134C97 |

102031

Chain ID

16/16

Tests passing

~15s

Detect → Settle

4

Contracts live

# We found a real constraint in Creditcoin's precompile — and fixed the architecture around it.

Testing against the real chain (not assumption) showed the Attestcoin Block Prover only answers a transaction where it's the direct, top-level target — never a call nested inside another contract.

## ✗ Nested call

Contract → precompile, from inside another contract's execution. Fails at the call boundary — never reaches verification logic.

## ✓ Top-level call

Oracle → precompile directly, as its own transaction. Reaches real verification logic — accepts or rejects for real.

Fix: verification now happens off-chain, by each independent oracle, backed by a public, checkable transaction hash — not a cosmetic workaround, a real architectural correction.

# What's left, and why it's not on us

- **Real satellite telemetry from Spacecoin** — their own docs confirm it isn't published on-chain today, on any chain. Outreach sent; no reply yet.
- **A real Spacecoin transaction** to build a genuine Attestcoin proof from — the proof *shape* is real, the content is still fabricated pending this.
- A product decision on whether coverage should also check Spacecoin's real payment activity.
- Security audit and legal review before any of this touches real funds.

Everything downstream of telemetry — proof, verification, settlement, payout — is built, tested, and live today.



SEE IT YOURSELF

# Try it live

## Live app

[space-shield-sigma.vercel.app](https://space-shield-sigma.vercel.app)

Connect a wallet, lock coverage, trigger a real outage on Creditcoin testnet, watch the real precompile respond.

## Source code

[github.com/midexol/SpaceShield](https://github.com/midexol/SpaceShield)

Every claim in this deck is reproducible from the repo — contracts, tests, oracle worker, architecture decision log.

THANK YOU

**Detection, proof, and payout —  
with no human in the loop.**

That's *SpaceShield*.